O'Neill §1.1 Problem Set

Exercise 1. *Let $f = x^2y$ and $g = y\sin z$ be functions on $\mathbf{E}^3$. Express the following functions in terms of x, y, z:*

(a) fg^2

(b) $\frac{\partial f}{\partial x}g + \frac{\partial g}{\partial y}f$

(c) $\frac{\partial^2(fg)}{\partial y \partial z}$

(d) $\frac{\partial}{\partial y}(\sin f)$

Solution. (a) Substituting the functions directly:

$$fg^2 = (x^2y)(y\sin z)^2 = x^2y \cdot y^2 \sin^2 z = x^2y^3 \sin^2 z.$$

(b) First, we compute the partial derivatives:

$$\frac{\partial f}{\partial x} = 2xy, \quad \frac{\partial g}{\partial y} = \sin z.$$

Substituting these back into the expression yields:

$$\frac{\partial f}{\partial x}g + \frac{\partial g}{\partial y}f = (2xy)(y\sin z) + (\sin z)(x^2y) = 2xy^2\sin z + x^2y\sin z = xy(2y+x)\sin z.$$

(c) First compute the product $fg = (x^2y)(y\sin z) = x^2y^2\sin z$. Differentiating with respect to z:

$$\frac{\partial(fg)}{\partial z} = x^2y^2\cos z.$$

Now differentiating this result with respect to y:

$$\frac{\partial^2(fg)}{\partial y \partial z} = \frac{\partial}{\partial y}(x^2y^2\cos z) = 2x^2y\cos z.$$

(d) Applying the chain rule with respect to y:

$$\frac{\partial}{\partial y}(\sin f) = \cos(f) \cdot \frac{\partial f}{\partial y} = \cos(x^2y) \cdot (x^2) = x^2\cos(x^2y).$$

□

Exercise 2. *Find the value of the function $f = x^2y - y^2z$ at each point:*

(a) $(1, 1, 1)$

(b) $(3, -1, \frac{1}{2})$

(c) $(a, 1, 1-a)$

(d) (t, t^2, t^3)

Solution. We substitute the coordinates of each point into $f(x, y, z) = x^2y - y^2z$:

(a) For $(1, 1, 1)$:
$$f(1, 1, 1) = (1)^2(1) - (1)^2(1) = 1 - 1 = 0.$$

(b) For $(3, -1, \frac{1}{2})$:
$$f(3, -1, \tfrac{1}{2}) = (3)^2(-1) - (-1)^2\left(\tfrac{1}{2}\right) = -9 - \tfrac{1}{2} = -\tfrac{19}{2}.$$

(c) For $(a, 1, 1-a)$:
$$f(a, 1, 1-a) = (a)^2(1) - (1)^2(1-a) = a^2 - 1 + a = a^2 + a - 1.$$

(d) For (t, t^2, t^3):
$$f(t, t^2, t^3) = (t)^2(t^2) - (t^2)^2(t^3) = t^4 - t^4 \cdot t^3 = t^4 - t^7.$$

□

Exercise 3. *Express $\partial f/\partial x$ in terms of $x, y,$ and z if*

(a) $f = x\sin(xy) + y\cos(xz)$.

(b) $f = \sin g, \quad g = e^h, \quad h = x^2 + y^2 + z^2$.

Solution. (a) Applying the product rule to the first term and the chain rule to both terms:
$$\begin{aligned}\frac{\partial f}{\partial x} &= \frac{\partial}{\partial x}[x] \cdot \sin(xy) + x \cdot \frac{\partial}{\partial x}[\sin(xy)] + y \cdot \frac{\partial}{\partial x}[\cos(xz)] \\ &= 1 \cdot \sin(xy) + x \cdot \big(y\cos(xy)\big) + y \cdot \big(-z\sin(xz)\big) \\ &= \sin(xy) + xy\cos(xy) - yz\sin(xz).\end{aligned}$$

(b) By composite substitution, $f = \sin\left(e^{x^2+y^2+z^2}\right)$. Applying the chain rule successively:
$$\begin{aligned}\frac{\partial f}{\partial x} &= \cos(g) \cdot \frac{\partial g}{\partial x} = \cos\left(e^h\right) \cdot e^h \frac{\partial h}{\partial x} \\ &= \cos\left(e^{x^2+y^2+z^2}\right) \cdot e^{x^2+y^2+z^2} \cdot (2x) \\ &= 2xe^{x^2+y^2+z^2}\cos\left(e^{x^2+y^2+z^2}\right).\end{aligned}$$

□

Exercise 4. *If g_1, g_2, g_3 and h are real-valued functions on $\mathbf{E}^3$, then $f = h(g_1, g_2, g_3)$ is the function such that $f(\mathbf{p}) = h(g_1(\mathbf{p}), g_2(\mathbf{p}), g_3(\mathbf{p}))$ for all $\mathbf{p}$. Express $\partial f/\partial x$ in terms of $x, y,$ and z, if $h = x^2 - yz$ and*

(a) $f = h(x + y, y^2, x + z)$.

(b) $f = h(e^x, e^{x+y}, e^z)$.

(c) $f = h(x, -x, x)$.

Solution. Let $h(u, v, w) = u^2 - vw$. By the multi-variable chain rule:

$$\frac{\partial f}{\partial x} = \frac{\partial h}{\partial u}\frac{\partial g_1}{\partial x} + \frac{\partial h}{\partial v}\frac{\partial g_2}{\partial x} + \frac{\partial h}{\partial w}\frac{\partial g_3}{\partial x} = 2u\frac{\partial g_1}{\partial x} - w\frac{\partial g_2}{\partial x} - v\frac{\partial g_3}{\partial x}.$$

(a) Here $g_1 = x + y$, $g_2 = y^2$, and $g_3 = x + z$. The inner partial derivatives are:

$$\frac{\partial g_1}{\partial x} = 1, \quad \frac{\partial g_2}{\partial x} = 0, \quad \frac{\partial g_3}{\partial x} = 1.$$

Substituting these along with $u = x + y$, $v = y^2$, and $w = x + z$:

$$\frac{\partial f}{\partial x} = 2(x + y)(1) - (x + z)(0) - (y^2)(1) = 2x + 2y - y^2.$$

(b) Here $g_1 = e^x$, $g_2 = e^{x+y}$, and $g_3 = e^z$. The inner partial derivatives are:

$$\frac{\partial g_1}{\partial x} = e^x, \quad \frac{\partial g_2}{\partial x} = e^{x+y}, \quad \frac{\partial g_3}{\partial x} = 0.$$

Substituting into the chain rule form:

$$\frac{\partial f}{\partial x} = 2(e^x)(e^x) - (e^z)(e^{x+y}) - (e^{x+y})(0) = 2e^{2x} - e^{x+y+z}.$$

(c) Here $g_1 = x$, $g_2 = -x$, and $g_3 = x$. The inner partial derivatives are:

$$\frac{\partial g_1}{\partial x} = 1, \quad \frac{\partial g_2}{\partial x} = -1, \quad \frac{\partial g_3}{\partial x} = 1.$$

Substituting into the chain rule form:

$$\frac{\partial f}{\partial x} = 2(x)(1) - (x)(-1) - (-x)(1) = 2x + x + x = 4x.$$

$\square$